Maciej Tkacz

Simulation Engineer - Physics-Based Robot Simulation, CAD-to-Sim, Python

Profile

Simulation engineer with two years of commercial experience building physics-based robot simulations and digital twins of production systems, scripting simulation logic in Python, and turning CAD models into working simulation assets. Mechanical Engineering student at Cracow University of Technology, so the physical parameters, kinematics and constraints I tune in a simulator are quantities I understand from first principles rather than numbers I adjust until the scene looks right. Currently a Robotics Engineer at ASTOR, programming Kawasaki and Epson industrial robots and their vision systems, building digital twins of production lines and commissioning the physical stations those twins describe - so I routinely see where a simulation and the real machine disagree. I also designed, built and programmed a SCARA robotic arm from scratch, including its forward and inverse kinematics, and validated the control software against the physical arm. I am now moving this work onto the OpenUSD and Isaac Sim pipeline, which is the toolchain rather than the discipline that is new to me.

Experience

2026-07 – present

Robotics Engineer

ASTOR, Krakow, Poland

- Build digital twins of production lines and robot simulations, validating reach, motion sequences and process behaviour before physical implementation.
- Program Kawasaki and Epson robots, including vision-guided applications: calibration, object models and search setup, part queues, and several different objects on a feeder at the same time.
- Run the Epson vision loop inside the RC+ simulator as well as on the station, and do the same class of vision work on the Kawasaki vision system.
- Design, build and commission demonstration stations presenting these robots in different applications - mechanical assembly, hardware-software integration, programming and bring-up.
- Program Astraada controllers in CODESYS: PLC logic, HMI screens, and the cell network on Ethernet/IP, PROFINET, Modbus and EtherCAT.
- For client projects, test a robot configuration in simulation first, then build a physical test station around the real robot and repeat the trial, to see how that configuration behaves in the actual application. Write the report with the measurements and the full procedure.

2025-01 – 2026-05

Application Engineer

AIAutomation, Poland

- Built robotics simulations and digital twins of production workcells in Visual Components for automotive clients.
- Wrote Python scripts, plugins and functions inside Visual Components: automation and process logic where the built-in options were enough, and custom behaviour where they were not.
- Developed robot paths, motion sequences, process logic, collision checking and cycle flows; validated processes virtually before implementation.
- Prepared and optimized workcell layouts and 3D geometry for simulation use, working from CAD in SolidWorks and Autodesk Inventor.
- Worked from customer documentation and engineering standards, and presented simulation assumptions, constraints and improvement proposals directly to client engineers.

2023-06 – 2023-08

3D Printing and CAD Design Specialist

Cubic Inch Additive Manufacturing, Piaseczno, Poland

- Designed and optimized CAD models in Fusion 360 and Autodesk Inventor for additive manufacturing.
- Ran FDM, MJF and SLA printers in production: process parameters, post-processing and quality control.
- Coordinated production tasks in a 10-person team, balancing quality, manufacturability and deadlines.

2022-05

Robotics and 3D Printing Intern

ASTOR Robotics Center, Krakow, Poland

- Assembled mechanical hardware and 3D-printed components for Kawasaki robots and Astorino robot platforms.
- Programmed Kawasaki robots via teach pendant and wrote basic motion programs.



Contact

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Krakow, Poland
EU citizen, no sponsorship

Languages

Polish - native
English - C1

Simulation

Visual Components
Python plugins in VC
Digital twins
Physics-based simulation
Isaac Sim (in progress)

Vision

Epson vision in RC+
Kawasaki vision
Calibration and search
Queues, multi-part feeder

Controls

CODESYS, Astraada PLC
HMI
Ethernet/IP, PROFINET
Modbus, EtherCAT

Code

Python
C++
Git

CAD

SolidWorks
Inventor
Fusion 360
AutoCAD

Robots

Kawasaki
Epson
Self-built SCARA

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Maciej Tkacz

Selected project

Personal – R&D

SCARA Robotic Arm - designed, built, programmed and measured

Personal R&D project

- Designed the complete machine: mechanical concept, CAD, kinematic layout and gear ratios, selection of NEMA stepper actuators and TMC/DRV drivers, microcontroller electronics and wiring.
- Implemented forward and inverse kinematics in Python and C++ and validated the control software against the physical arm rather than only against the model.
- Designed every printed part for stiffness and assembly in PET-G and carbon-fibre reinforced material, iterating through repeated print, assemble and test cycles.
- The reason this matters for a simulation role: I own both halves of the comparison. I have the CAD, the real masses and inertias, the kinematic model and the physical machine, which makes a measured sim-to-real study possible rather than hypothetical.

Education

2024-10 – present

Mechanical Engineering, Engineer's degree

Cracow University of Technology, Krakow

- In progress. Coursework: dynamic systems modelling, automation and control, analytical mechanics, mechatronics.

PKMechPower, Mechanical Section

Student research group, Cracow University of Technology

- CAD and 3D-printed parts for a student vehicle, optimized for mass and strength.

2019-09 – 2024-04

Software Technician

Zespol Szkol Informatycznych, Kielce

Certificates

2022-05

Kawasaki robot operation and programming

Integrator course with certificate, ASTOR Robotics Center.

Epson industrial robot programming

Certified course.

Python

Courses, plus self-directed Python and C++ in robotics and simulation projects.

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